EED 101 INTRODUCTION TO ELECTRICAL ENGINEERING

SPRING 2017 **INSTRUCTOR: DR G NAVEEN BABU**

PROBLEM SET 2 SOLUTIONS

REFERENCE TEXT BOOK: INTRODUCTORY CIRCUIT ANALYSIS, ROBERT BOYLESTAD, ***12TH EDITION***

Chapter 6: 8, 16, 26, 32, 42

Chapter 7: 2c, 8, 22, 29, 31

Chapter 6:

8,

$$24\ \Omega \parallel 24\ \Omega = 12\ \Omega$$

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{12\ \Omega} + \frac{1}{120\ \Omega}$$

$$0.1\ \text{S} = \frac{1}{R_1} + 0.08333\ \text{S} + 0.00833\ \text{S}$$

$$0.1\ \text{S} = \frac{1}{R_1} + 0.09167\ \text{S}$$

$$\frac{1}{R_1} = 0.1\ \text{S} - 0.09167\ \text{S} = 0.00833\ \text{S}$$

$$R_1 = \frac{1}{0.00833\ \text{S}} = \mathbf{120\ \Omega}$$

16,

a. $V = \mathbf{48\ V}$

b. $I_2 = \dfrac{48\ \text{V}}{18\ \text{k}\Omega} = \mathbf{2.67\ mA}$

c. $I_s = \dfrac{48\ \text{V}}{3\ \text{k}\Omega} + \dfrac{48\ \text{V}}{12\ \text{k}\Omega} + I_2 = 16\ \text{mA} + 4\ \text{mA} + 2.67\ \text{mA} = \mathbf{22.67\ mA}$

d. $P = \dfrac{V^2}{R} = \dfrac{E^2}{R} = \dfrac{(48\ \text{V})^2}{12\ \text{k}\Omega} = \mathbf{192\ mW}$

26,

a. $\sum I_i = \sum I_o$

$8\ \text{A} = 3\ \text{A} + I_2$

$I_2 = 8\ \text{A} - 3\ \text{A} = \mathbf{5\ A}$, $I_3 = \mathbf{3\ A}$

$\sum I_i = \sum I_o$

$I_2 + I_3 = I_4$

$I_4 = 5\ \text{A} + 3\ \text{A} = \mathbf{8\ A}$

b. $\sum I_i = \sum I_o$

$I_s = 36\ \text{mA} + 4\ \text{mA} = \mathbf{40\ mA}$

$\sum I_i = \sum I_o$

$36\ \text{mA} = I_3 + 20\ \text{mA}$

$I_3 = 36\ \text{mA} - 20\ \text{mA} = \mathbf{16\ mA}$

$\sum I_i = \sum I_o$

$4\ \text{mA} + 20\ \text{mA} = I_4$

$I_4 = \mathbf{24\ mA}$

$I_5 = I_s = \mathbf{40\ mA}$

32,

a. $I_1 = \dfrac{8\text{ k}\Omega(20\text{ mA})}{2\text{ k}\Omega + 8\text{ k}\Omega} = \mathbf{16\ mA}$

$I_2 = 20\text{ mA} - 16\text{ mA} = \mathbf{4\ mA}$

b. $I_{2.4\text{k}\Omega} = 2.5\text{ A} = \dfrac{1\text{ k}\Omega(I_T)}{1\text{ k}\Omega + 2.4\text{ k}\Omega} = \dfrac{1\text{ k}\Omega(I_T)}{3.4\text{ k}\Omega}$

and $I_T = \dfrac{3.4\text{ k}\Omega(2.5\text{ A})}{1\text{ k}\Omega} = \mathbf{8.5\ A}$

$I_1 = I_T - 2.5\text{ A} = 8.5\text{ A} - 2.5\text{ A} = \mathbf{6\ A}$

42

a. $I_1 = \dfrac{20\text{ V}}{4\ \Omega} = \mathbf{5\ A},\ I_2 = \mathbf{0\ A}$

b. $V_1 = \mathbf{0\ V},\ V_2 = \mathbf{20\ V}$

c. $I_s = I_1 = \mathbf{5\ A}$

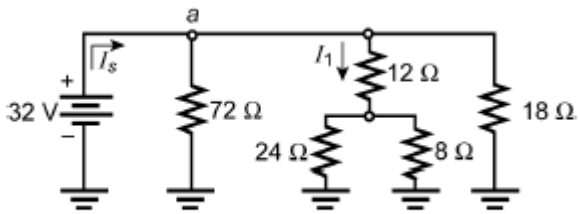
Chapter 7:

2c,

$R_T = \dfrac{4\ \Omega}{2} + 10\ \Omega = 2\ \Omega + 10\ \Omega = \mathbf{12\ \Omega}$

8,

Redrawn:



a. $V_a = \mathbf{32\ V}$

$8\ \Omega \parallel 24\ \Omega = 6\ \Omega$

$V_b = \dfrac{6\ \Omega(32\text{ V})}{6\ \Omega + 12\ \Omega} = \mathbf{10.67\ V}$

b. $I_1 = \dfrac{32\text{ V}}{12\ \Omega + 6\ \Omega} = \dfrac{32\text{ V}}{18\ \Omega} = \mathbf{1.78\ A}$

$R_T = 72\ \Omega \parallel \underbrace{18\ \Omega \parallel 18\ \Omega}_{9\ \Omega} = 8.12\ \Omega$

$$I_s = \frac{E}{R_T} = \frac{32\text{ V}}{8.12\ \Omega} = \mathbf{3.94\ A}$$

22,

a. Applying Kirchoff's voltage law in the CCW direction in the upper "window":

$$+18\text{ V} + 20\text{ V} - V_{8\Omega} = 0$$
$$V_{8\Omega} = 38\text{ V}$$
$$I_{8\Omega} = \frac{38\text{ V}}{8\ \Omega} = 4.75\text{ A}$$
$$I_{3\Omega} = \frac{18\text{ V}}{3\ \Omega + 6\ \Omega} = \frac{18\text{ V}}{9\ \Omega} = 2\text{ A}$$

KCL: $I_{18V} = 4.75\text{ A} + 2\text{ A} = \mathbf{6.75\ A}$

b. $V = (I_{3\Omega})(6\ \Omega) + 20\text{ V} = (2\text{ A})(6\ \Omega) + 20\text{ V} = 12\text{ V} + 20\text{ V} = \mathbf{32\ V}$

29,

a. $R_{10} + R_{11} \parallel R_{12} = 1\ \Omega + 2\ \Omega \parallel 2\ \Omega = 2\ \Omega$
$R_4 \parallel (R_5 + R_6) = 10\ \Omega \parallel 10\ \Omega = 5\ \Omega$
$R_1 + R_2 \parallel (R_3 + 5\ \Omega) = 3\ \Omega + 6\ \Omega \parallel 6\ \Omega = 6\ \Omega$
$R_T = 2\ \Omega \parallel 3\ \Omega \parallel 6\ \Omega = 2\ \Omega \parallel 2\ \Omega = 1\ \Omega$
$I = 12\text{ V}/1\ \Omega = \mathbf{12\ A}$

b. $I_1 = 12\text{ V}/6\ \Omega = 2\text{ A}$

$$I_3 = \frac{6\ \Omega(2\text{ A})}{6\ \Omega + 6\ \Omega} = 1\text{ A}$$

$$I_4 = \frac{1\text{ A}}{2} = \mathbf{0.5\ A}$$

c. $I_6 = I_4 = \mathbf{0.5\ A}$

d. $I_{10} = \dfrac{12\text{ A}}{2} = \mathbf{6\ A}$

31

a. $R_{10} + R_{11} \parallel R_{12} = 1\ \Omega + 2\ \Omega \parallel 2\ \Omega = 2\ \Omega$

$R_4 \parallel (R_5 + R_6) = 10\ \Omega \parallel 10\ \Omega = 5\ \Omega$

$R_1 + R_2 \parallel (R_3 + 5\ \Omega) = 3\ \Omega + 6\ \Omega \parallel 6\ \Omega = 6\ \Omega$

$R_T = 2\ \Omega \parallel 3\ \Omega \parallel 6\ \Omega = 2\ \Omega \parallel 2\ \Omega = 1\ \Omega$

$I = 12\text{ V}/1\ \Omega = \mathbf{12\ A}$

b. $I_1 = 12\text{ V}/6\ \Omega = 2\text{ A}$

$$I_3 = \frac{6\ \Omega(2\text{ A})}{6\ \Omega + 6\ \Omega} = 1\text{ A}$$

$$I_4 = \frac{1\text{ A}}{2} = \mathbf{0.5\ A}$$

c. $I_6 = I_4 = \mathbf{0.5\ A}$

d. $$I_{10} = \frac{12\text{ A}}{2} = \mathbf{6\ A}$$

$I_{R_1} = 40\text{ mA}$

$I_{R_2} = 40\text{ mA} - 10\text{ mA} = 30\text{ mA}$

$I_{R_3} = 30\text{ mA} - 20\text{ mA} = 10\text{ mA}$

$I_{R_5} = 40\text{ mA}$

$I_{R_4} = 40\text{ mA} - 4\text{ mA} = 36\text{ mA}$

$$R_1 = \frac{V_{R_1}}{I_{R_1}} = \frac{120\text{ V} - 100\text{ V}}{40\text{ mA}} = \frac{20\text{ V}}{40\text{ mA}} = \mathbf{0.5\ k\Omega}$$

$$R_2 = \frac{V_{R_2}}{I_{R_2}} = \frac{100\text{ V} - 40\text{ V}}{30\text{ mA}} = \frac{60\text{ V}}{30\text{ mA}} = \mathbf{2\ k\Omega}$$

$$R_3 = \frac{V_{R_3}}{I_{R_3}} = \frac{40\text{ V}}{10\text{ mA}} = \mathbf{4\ k\Omega}$$

$$R_4 = \frac{V_{R_4}}{I_{R_4}} = \frac{36\text{ V}}{36\text{ mA}} = \mathbf{1\ k\Omega}$$

$$R_5 = \frac{V_{R_5}}{I_{R_5}} = \frac{60\text{ V} - 36\text{ V}}{40\text{ mA}} = \frac{24\text{ V}}{40\text{ mA}} = \mathbf{0.6\ k\Omega}$$

$P_1 = I_1^2 R_1 = (40\text{ mA})^2 0.5\text{ k}\Omega = \mathbf{0.8\ W}$ (1 watt resistor)

$P_2 = I_2^2 R_2 = (30\text{ mA})^2 2\text{ k}\Omega = \mathbf{1.8\ W}$ (2 watt resistor)

$P_3 = I_3^2 R_3 = (10\text{ mA})^2 4\text{ k}\Omega = \mathbf{0.4\ W}$ (1/2 watt or 1 watt resistor)

$P_4 = I_4^2 R_4 = (36\text{ mA})^2 1\text{ k}\Omega = \mathbf{1.3\ W}$ (2 watt resistor)

$P_5 = I_5^2 R_5 = (40\text{ mA})^2 0.6\text{ k}\Omega = \mathbf{0.96\ W}$ (1 watt resistor)

All power levels less than **2 W**. Four less than **1 W**.